Experiment 2: Covariance, Eigenvalues and Eigenvectors Analysis

Aim

To standardise patient data, compute covariance and correlation matrices, determine eigenvalues and eigenvectors, and analyse the percentage of variance explained by each principal direction.

Mathematical Formulae

1. Standardisation

$$z = \frac{x - \mu}{\sigma}$$

2. Covariance

$$\mathrm{Cov}(X, Y) = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

3. Correlation

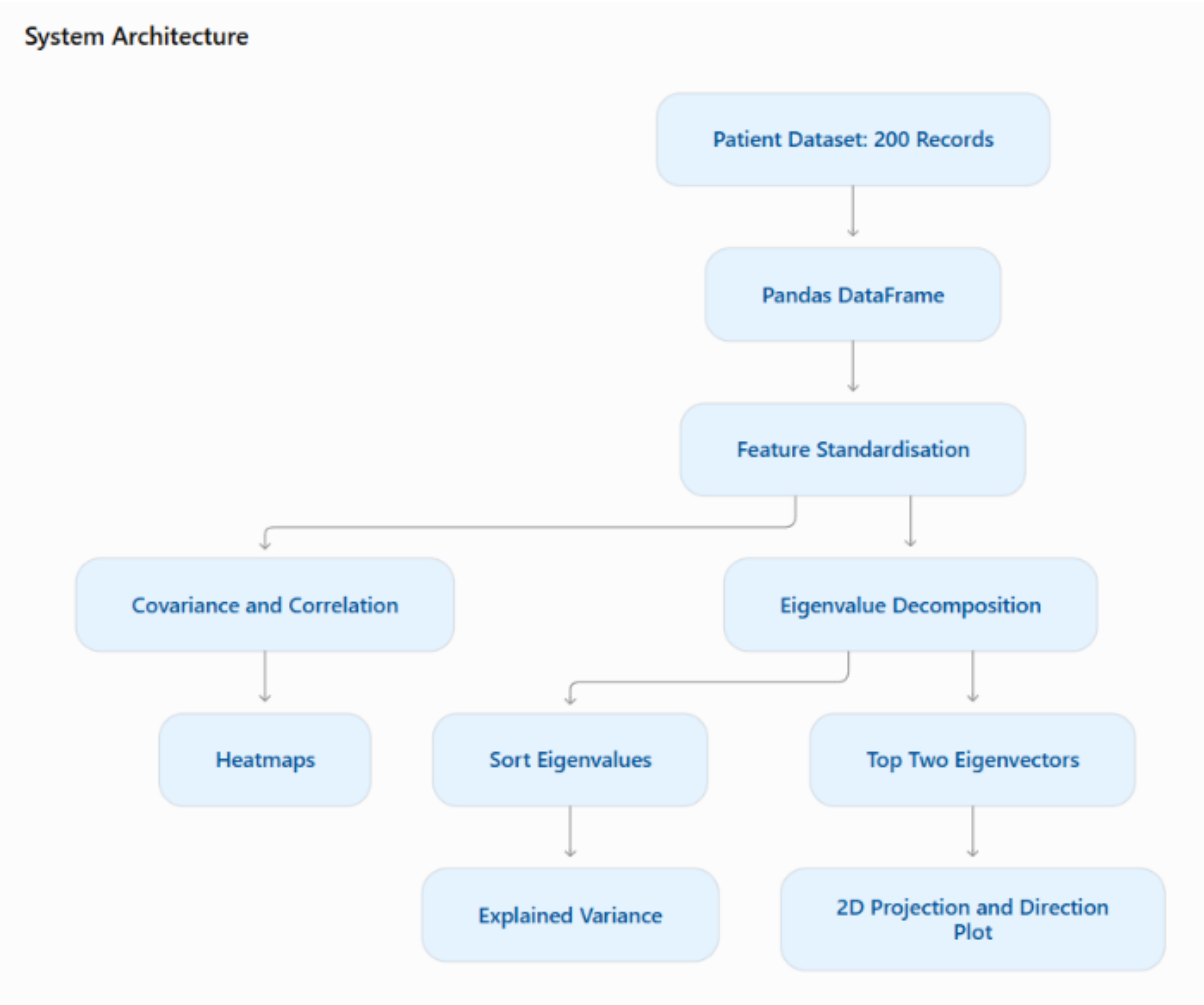
$$r_{XY} = \frac{\mathrm{Cov}(X, Y)}{s_X s_Y}$$

4. Eigenvalue equation

$$Av = \lambda v$$

5. Explained variance

$$EV_i = \frac{\lambda_i}{\sum_j \lambda_j} \times 100$$



Algorithm / Procedure

1. Generate or load 200 patient records containing five health features.
2. Standardise all features using the Z-score formula.
3. Calculate covariance and correlation matrices.
4. Plot the two matrices as heatmaps.
5. Compute eigenvalues and eigenvectors using NumPy.
6. Sort the eigenvalues in descending order.
7. Calculate the explained variance percentage.
8. Project the patient records onto the top two eigenvectors and visualise the results.

Python Program

```
import numpy as np
```

```
import pandas as pd
import matplotlib.pyplot as plt

np.random.seed(42)

# 1. Generate synthetic patient data (200 records)
n = 200

health = np.random.normal(0, 1, n)
haemoglobin = 13.5 + 1.2 * health + np.random.normal(0, 0.5, n)
rbc = 4.8 + 0.3 * health + np.random.normal(0, 0.15, n)
glucose = 100 + np.random.normal(0, 12, n)
cholesterol = 190 + 12 * health + np.random.normal(0, 15, n)
bmi = 25 + 2 * health + np.random.normal(0, 2, n)

df = pd.DataFrame({
  "Haemoglobin": haemoglobin,
  "RBC": rbc,
  "Glucose": glucose,
  "Cholesterol": cholesterol,
  "BMI": bmi
})

print("First 5 Patient Records:\n", df.head())

# 2. Standardise each feature (Z-score)
X = (df - df.mean()) / df.std(ddof=1)

print("\nStandardised Data:\n", X.head())

# 3. Covariance and correlation matrices
cov_matrix = X.cov()
corr_matrix = X.corr()
```

```
print("\nCovariance Matrix:\n", cov_matrix)
print("\nCorrelation Matrix:\n", corr_matrix)

# Plot covariance and correlation heatmaps
fig, axes = plt.subplots(1, 2, figsize=(13, 5))

for ax, matrix, title in [
  (axes[0], cov_matrix, "Covariance Matrix"),
  (axes[1], corr_matrix, "Correlation Matrix")
]:
  im = ax.imshow(matrix, cmap="coolwarm", vmin=-1, vmax=1)
  ax.set_xticks(range(5), matrix.columns, rotation=45, ha="right")
  ax.set_yticks(range(5), matrix.index)
  ax.set_title(title)

  for i in range(5):
    for j in range(5):
      ax.text(j, i, f"{matrix.iloc[i, j]:.2f}",
          ha="center", va="center", fontsize=8)

fig.colorbar(im, ax=axes, shrink=0.8)
plt.tight_layout()
plt.show()

# 4. Eigenvalues and eigenvectors
values, vectors = np.linalg.eigh(cov_matrix.values)

# Sort in descending order
order = np.argsort(values)[::-1]
values = values[order]
vectors = vectors[:, order]

print("\nEigenvalues:\n", values)
print("\nEigenvectors (columns):\n", vectors)
```

```
# 5. Percentage of variance explained
explained = values / values.sum() * 100
print("\nExplained Variance (%):\n", np.round(explained, 2))
print("Top 2 Components Explain:",
    round(explained[:2].sum(), 2), "%")

# 6. Project standardised data onto top two eigenvectors
projected = X.values @ vectors[:, :2]

plt.figure(figsize=(7, 5))
plt.scatter(projected[:, 0], projected[:, 1], alpha=0.6)
plt.title("Patient Data Projected onto Top Two Eigenvectors")
plt.xlabel("First Principal Direction")
plt.ylabel("Second Principal Direction")
plt.grid(True)
plt.tight_layout()
plt.show()

# 7. Plot the top two eigenvectors as arrows
# Coordinates are the first two original standardised features
plt.figure(figsize=(7, 5))
plt.scatter(X.iloc[:, 0], X.iloc[:, 1], alpha=0.3)

for i in range(2):
  plt.arrow(0, 0,
        vectors[0, i] * 2,
        vectors[1, i] * 2,
        head_width=0.08, length_includes_head=True)
  plt.text(vectors[0, i] * 2.1,
       vectors[1, i] * 2.1,
       f"PC{i+1}")

plt.title("Top Eigenvector Directions")
```

```
plt.xlabel("Standardised Haemoglobin")

plt.ylabel("Standardised RBC")

plt.grid(True)

plt.tight_layout()

plt.show()
```

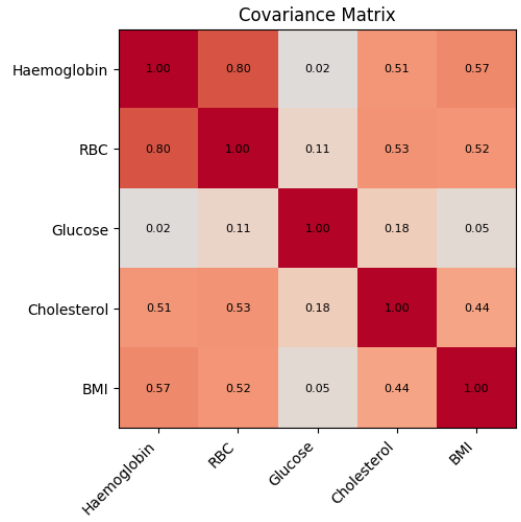
Output

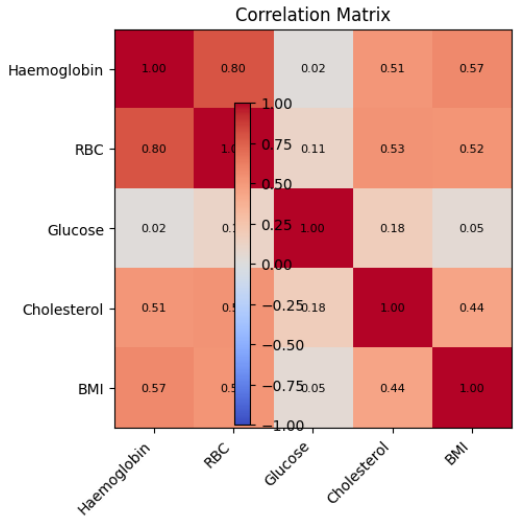
```
First 5 Patient Records:
    Haemoglobin       RBC     Glucose  Cholesterol        BMI
0    14.274951  4.709850  109.083863   210.034827  28.792139
1    13.614475  4.668614   88.934016   180.600157  26.572739
2    14.818752  4.995093  110.435271   199.214074  26.414638
3    15.854537  5.263956  116.267654   201.342229  26.752186
4    12.530181  4.662244  104.961219   180.672716  25.928140

Standardised Data:
    Haemoglobin       RBC   Glucose  Cholesterol       BMI
0     0.618044 -0.218037  0.733661     1.039101  1.361631
1     0.095338 -0.356210 -0.913256    -0.605259  0.523679
2     1.048413  0.737753  0.844116     0.434603  0.463986
3     1.868143  1.638658  1.320817     0.553492  0.591430
4    -0.762782 -0.377555  0.396703    -0.601205  0.280305

Covariance Matrix:
             Haemoglobin       RBC   Glucose  Cholesterol       BMI
Haemoglobin     1.000000  0.796604  0.018221     0.508483  0.573272
RBC             0.796604  1.000000  0.114144     0.529212  0.522586
Glucose         0.018221  0.114144  1.000000     0.177925  0.048671
Cholesterol     0.508483  0.529212  0.177925     1.000000  0.444871
BMI             0.573272  0.522586  0.048671     0.444871  1.000000

Correlation Matrix:
             Haemoglobin       RBC   Glucose  Cholesterol       BMI
Haemoglobin     1.000000  0.796604  0.018221     0.508483  0.573272
RBC             0.796604  1.000000  0.114144     0.529212  0.522586
Glucose         0.018221  0.114144  1.000000     0.177925  0.048671
Cholesterol     0.508483  0.529212  0.177925     1.000000  0.444871
BMI             0.573272  0.522586  0.048671     0.444871  1.000000
/tmp/ipykernel_792/1013122075.py:57: UserWarning: This figure includes Axes that are not compatible with tight_layout, so results might be incorrect.
  plt.tight_layout()
```

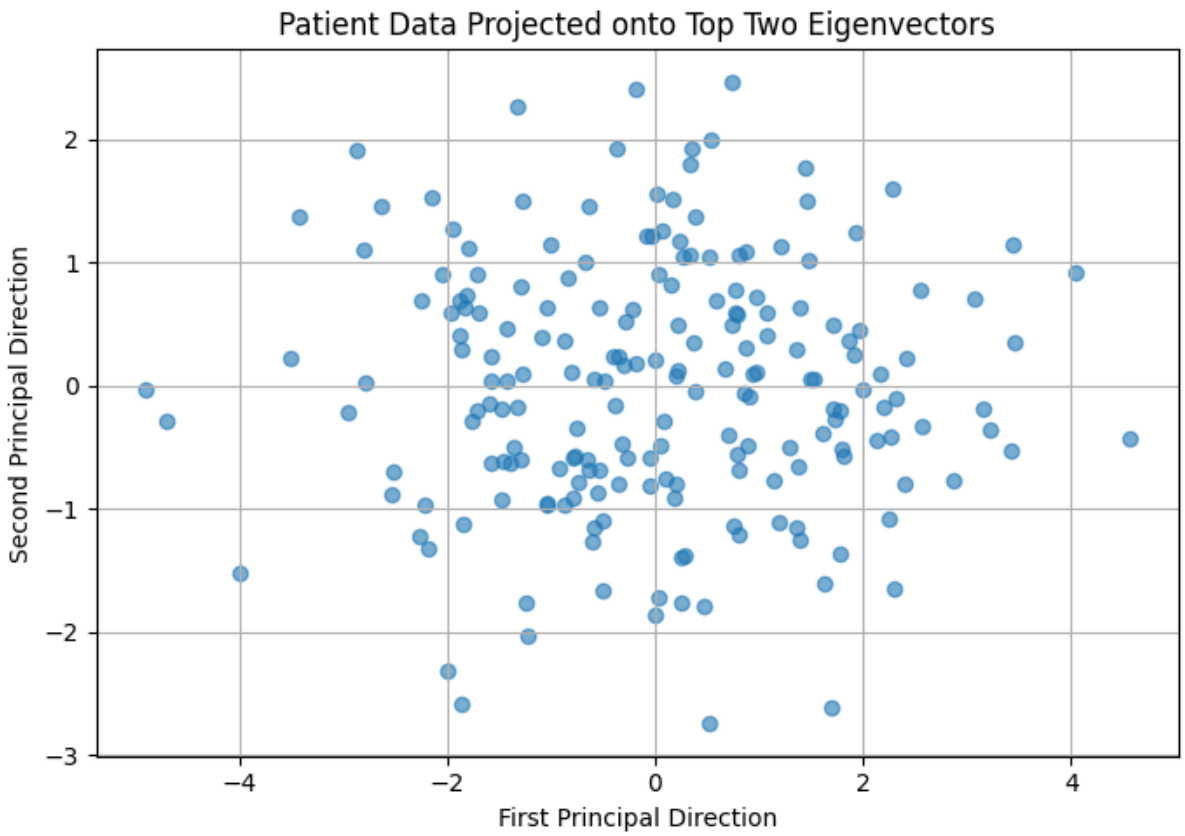





```
Eigenvalues:
 [2.71982553 1.01669601 0.53969303 0.52943591 0.19434951]

Eigenvectors (columns):
 [[-0.53593344  0.17418483  0.23511038  0.32882589 -0.7204385 ]
 [-0.53422759  0.04897168  0.20820548  0.45027913  0.68271651]
 [-0.10099182 -0.96075522  0.24546495 -0.00709272 -0.08029139]
 [-0.452508   -0.17235456 -0.87113522 -0.07770728 -0.02480777]
 [-0.46088327  0.12043556  0.28678212 -0.82645841  0.08834249]]

Explained Variance (%):
 [54.4  20.33 10.79 10.59  3.89]
Top 2 Components Explain: 74.73 %
```



Result

Thus, the patient features were standardised and analysed using covariance, correlation and eigenvalue decomposition. The principal directions of variation and their explained variance percentages were calculated and visualised.